

PHYSICS FORMULA SHEET

PROPERTY OF TUTOR SYNERGIST (NOT FOR SALE)

For questions regarding any of these formulae, contact directly: Phone: +234 7072 4774 WhatsApp:

<https://wa.link/y4sk57>

MECHANICS & MOTION

1. First equation of motion: $v = u + at$
2. Second equation of motion: $s = ut + \frac{1}{2}at^2$
3. Third equation of motion: $v^2 = u^2 + 2as$
4. Distance with average velocity: $s = (u + v)t / 2$
5. Newton's second law: $F = ma$
6. Weight: $W = mg$
7. Apparent weight in an elevator (accelerating up): $W' = m(g + a)$
8. Apparent weight in an elevator (accelerating down): $W' = m(g - a)$
9. Momentum: $p = mv$
10. Force from change in momentum: $F = (mv - mu) / t$
11. Impulse: $I = F \cdot t = mv - mu$
12. Law of conservation of momentum: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$
13. Newton's law of universal gravitation: $F = G \cdot m_1m_2 / r^2$
14. Acceleration due to gravity: $g = GM / r^2$
15. Velocity of free fall: $v = \sqrt{2gh}$
16. Escape velocity: $v_{esc} = \sqrt{2gR} = \sqrt{2GM / R}$
17. Period of a simple pendulum: $T = 2\pi\sqrt{l / g}$
18. Period of a spiral spring: $T = 2\pi\sqrt{m / k}$
19. Frequency: $f = 1 / T$
20. Angular velocity: $\omega = 2\pi f = 2\pi / T$
21. Linear velocity in circular motion: $v = r\omega$
22. Centripetal acceleration: $a = v^2 / r = r\omega^2$
23. Centripetal force: $F = mv^2 / r = mr\omega^2$
24. Moment of a force: $M = F \cdot d$
25. Principle of moments: Clockwise moments = Anticlockwise moments
26. Mechanical Advantage: $MA = \text{Load} / \text{Effort}$
27. Velocity Ratio: $VR = \text{Distance moved by effort} / \text{Distance moved by load}$
28. Efficiency of a machine: $\text{Efficiency} = (MA / VR) \times 100\%$
29. Velocity Ratio of an inclined plane: $VR = 1 / \sin \theta = l / h$
30. Velocity Ratio of a screw jack: $VR = 2\pi R / \text{pitch}$

31. Velocity Ratio of wheel and axle: $VR = \text{Radius of wheel} / \text{Radius of axle}$
32. Velocity Ratio of gear system: $VR = \text{Number of teeth on driven gear} / \text{Number of teeth on driver gear}$
33. Velocity Ratio of block and tackle (pulley): $VR = \text{Number of pulleys in the system}$

PROPERTIES OF MATTER, ELASTICITY & FLUIDS

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34. Density: $\rho = m / V$
35. Pressure: $P = F / A$
36. Pressure in a liquid: $P = h\rho g$
37. Total pressure in liquid: $P_{total} = P_{atmosphere} + h\rho g$
38. Upthrust (Archimedes' Principle): $F_B = V\rho g$
39. Relative Density: $RD = \rho_{substance} / \rho_{water}$
40. Relative Density using weight: $RD = \text{Weight in air} / \text{Upthrust in water}$
41. Relative Density of liquid: $RD = \text{Upthrust in liquid} / \text{Upthrust in water}$
42. Stress: $\text{Stress} = F / A$
43. Strain: $\text{Strain} = e / l$
44. Young's Modulus: $Y = \text{Stress} / \text{Strain} = Fl / Ae$
45. Hooke's Law: $F = ke$
46. Surface Tension: $\gamma = F / l$
47. Stokes' Law (Viscous force): $F = 6\pi\eta rv$
48. Coefficient of Viscosity: $\eta = Fl / Av$

THERMAL PHYSICS

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49. Specific heat capacity: $Q = mc\Delta\theta$
50. Heat capacity: $C = mc = Q / \Delta\theta$
51. Latent heat: $Q = mL$
52. Principle of heat exchange: $\text{Heat Lost} = \text{Heat Gained}$
53. Linear expansivity: $\alpha = \Delta l / (l_1\Delta\theta)$
54. Area expansivity: $\beta = \Delta A / (A_1\Delta\theta) = 2\alpha$

55. Volume expansivity: $\gamma = \Delta V / (V_1 \Delta \theta) = 3\alpha$
56. Real vs Apparent expansivity: $\gamma_{\text{real}} = \gamma_{\text{apparent}} + \gamma_{\text{vessel}}$
57. Rate of conduction heat transfer: $Q / t = (kA \Delta \theta) / l$
58. Stefan-Boltzmann Law (Radiation rate): $P = \sigma A T^4$
59. Kelvin temperature conversion: $K = ^\circ\text{C} + 273$
60. Boyle's Law: $P_1 V_1 = P_2 V_2$
61. Charles' Law: $V_1 / T_1 = V_2 / T_2$
62. Pressure (Gay-Lussac's) Law: $P_1 / T_1 = P_2 / T_2$
63. General Gas Law: $(P_1 V_1) / T_1 = (P_2 V_2) / T_2$
64. Ideal Gas Equation: $PV = nRT$
65. First Law of Thermodynamics: $\Delta U = Q - W$

WAVES, SOUND & OPTICS

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66. Wave speed: $v = f\lambda$
67. Wave period: $T = 1 / f$
68. Angular frequency: $\omega = 2\pi f = 2\pi / T$
69. Wave number: $k = 2\pi / \lambda$
70. Progressive wave equation (moving right): $y = A \sin(\omega t - kx) = A \sin[2\pi(ft - x/\lambda)]$
71. Progressive wave equation (moving left): $y = A \sin(\omega t + kx) = A \sin[2\pi(ft + x/\lambda)]$
72. Speed of transverse wave on string: $v = \sqrt{T / \mu}$
73. Fundamental frequency of string (Sonometer): $f_1 = (1 / 2l) \sqrt{T / \mu}$
74. Harmonics on a string / open pipe: $f_n = nf_1$ (where $n = 1, 2, 3, \dots$)
75. Harmonics in a closed pipe: $f_n = nf_1$ (where $n = 1, 3, 5, \dots$)
76. Fundamental frequency of closed pipe: $f_1 = v / 4l$
77. Fundamental frequency of open pipe: $f_1 = v / 2l$
78. Beat frequency: $f_{\text{beat}} = |f_1 - f_2|$
79. Doppler Effect (Observed frequency): $f' = f(v \pm v_{\text{observer}}) / (v \mp v_{\text{source}})$
80. Echo distance: $s = vt / 2$
81. Sound Intensity: $I = \text{Power} / \text{Area} = P / 4\pi r^2$

82. Sound intensity level (Decibels): $\beta = 10 \log_{10}(I / I_0)$
83. Snell's Law: $n = \sin i / \sin r = v_1 / v_2 = \lambda_1 / \lambda_2$
84. Refractive Index from depth: $n = \text{Real depth} / \text{Apparent depth}$
85. Critical angle: $\sin C = 1 / n$
86. Refractive Index of prism: $n = \sin[(A + D_m)/2] / \sin(A/2)$
87. Mirror and Lens formula: $1 / f = 1 / u + 1 / v$
88. Magnification: $m = v / u = \text{Image Height} / \text{Object Height}$
89. Power of a Lens: $P = 1 / f$ (f in meters)
90. Combined focal length of two thin lenses: $1 / F = 1 / f_1 + 1 / f_2$
91. Diffraction Grating equation: $d \sin \theta = n\lambda$

ELECTRICITY & MAGNETISM

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92. Electric Charge: $Q = It = ne$
93. Ohm's Law: $V = IR$
94. Resistance of a wire: $R = \rho l / A$
95. Resistors in Series: $R_{\text{total}} = R_1 + R_2 + R_3$
96. Resistors in Parallel: $1 / R_{\text{total}} = 1/R_1 + 1/R_2 + 1/R_3$
97. Two Resistors in Parallel: $R_{\text{total}} = (R_1 R_2) / (R_1 + R_2)$
98. EMF and Internal Resistance: $E = I(R + r) = V + Ir$
99. Internal Resistance: $r = R(E/V - 1)$
100. Coulomb's Law: $F = (1 / 4\pi \epsilon_0) \cdot (q_1 q_2 / r^2)$
101. Electric Field Strength: $E = F / q = V / d$
102. Electric Potential: $V = W / q = kq / r$
103. Capacitance: $C = Q / V$
104. Parallel Plate Capacitance: $C = \epsilon A / d$
105. Capacitors in Parallel: $C_{\text{total}} = C_1 + C_2 + C_3$

106. Capacitors in Series: $1 / C_{\text{total}} = 1/C_1 + 1/C_2 + 1/C_3$
107. Force on current-carrying conductor: $F = BIL \sin \theta$
108. Force on moving charge in magnetic field: $F = qvB \sin \theta$
109. Magnetic field of long straight wire: $B = \mu_0 I / 2\pi r$
110. Magnetic field inside a solenoid: $B = \mu_0 nI = \mu_0 NI / l$
111. Motional EMF: $E = Blv$
112. Faraday's Law of Induction: $E = -N(\Delta\Phi / \Delta t)$
113. Self-Inductance EMF: $E = -L(\Delta I / \Delta t)$
114. Magnetic Flux: $\Phi = BA \cos \theta$
115. Transformer Voltage and Turns Ratio: $V_p / V_s = N_p / N_s = I_s / I_p$
116. Transformer Efficiency: Efficiency = $(V_s I_s / V_p I_p) \times 100\%$
117. Inductive Reactance: $X_L = 2\pi fL$
118. Capacitive Reactance: $X_C = 1 / 2\pi fC$
119. Impedance in RLC Circuit: $Z = \sqrt{[R^2 + (X_L - X_C)^2]}$
120. Resonant Frequency in AC Circuit: $f_0 = 1 / (2\pi\sqrt{LC})$

ATOMIC & MODERN PHYSICS

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121. Photon Energy: $E = hf = hc / \lambda$
122. Speed of Light: $c = f\lambda$
123. Photoelectric Equation: $hf = W_0 + KE_{\text{max}} = hf_0 + \frac{1}{2}mv^2$
124. Stopping Potential: $eVs = KE_{\text{max}} = hf - W_0$
125. de Broglie Wavelength: $\lambda = h / p = h / mv$
126. Energy level transitions: $\Delta E = E_2 - E_1 = hf$
127. Rydberg Formula: $1 / \lambda = R_H(1/n_1^2 - 1/n_2^2)$
128. Hydrogen energy level formula: $E_n = -13.6 / n^2$ (in eV)
129. Mass-Energy Equivalence: $E = mc^2$

130. Radioactive Decay: $N = N_0(\frac{1}{2})^{(t / T_{1/2})}$
131. Decay Rate (Activity): $A = \lambda N$
132. Half-life relation to decay constant: $T_{1/2} = 0.693 / \lambda$
133. Nuclear Binding Energy: $E = \Delta m \cdot c^2$
134. Alpha Decay: ${}_Z^A X \rightarrow {}_{(Z-2)}^{A-4} Y + {}_2^4 \text{He}$
135. Beta Decay: ${}_Z^A X \rightarrow {}_{(Z+1)}^A Y + {}_{-1}^0 e$
136. Electron Capture: $p + e^- \rightarrow n + \nu$

FORMULAE FOR ENERGY

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137. Mechanical Work / Energy: $E = Fd \cos \theta$
138. Kinetic Energy (Linear): $KE = \frac{1}{2}mv^2 = p^2 / 2m$
139. Gravitational Potential Energy: $PE = mgh$
140. Universal Gravitational Potential Energy: $PE = -Gm_1m_2 / r$
141. Elastic Potential Energy (Spring/Wire): $E = \frac{1}{2}Fx = \frac{1}{2}kx^2$
142. Thermal / Heat Energy: $Q = mc\Delta\theta$
143. Latent Heat Energy: $Q = mL$
144. Electrical Energy: $E = Pt = IVt = I^2Rt = (V^2 / R)t$
145. Energy Stored in Capacitor: $E = \frac{1}{2}CV^2 = \frac{1}{2}QV = Q^2 / 2C$
146. Energy Stored in Inductor: $E = \frac{1}{2}LI^2$
147. Electrostatic Potential Energy: $E = qV = (1 / 4\pi\epsilon_0) \cdot (q_1q_2 / r)$
148. Photon Energy: $E = hf = hc / \lambda$
149. Mass-Energy Equivalence: $E = mc^2$

FORMULAE FOR POWER

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150. General Mechanical Power: $P = \text{Work} / \text{time} = E / t$

151. Power in Linear Motion: $P = \text{Force} \times \text{velocity} = Fv$
152. Electrical Power (General): $P = IV$
153. Electrical Power (Resistive): $P = I^2R$
154. Electrical Power (Voltage & Resistance): $P = V^2 / R$
155. Apparent Power in AC Circuit: $S = I_{\text{rms}} \cdot V_{\text{rms}}$
156. Real Power in AC Circuit: $P = I_{\text{rms}} \cdot V_{\text{rms}} \cdot \cos \phi$
157. Lens Power: $P = 1 / f$ (where f is in meters)